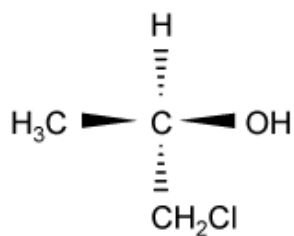
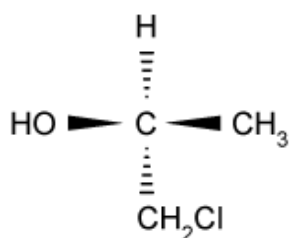


What structure is equivalent to Compound 1 (shown below)?



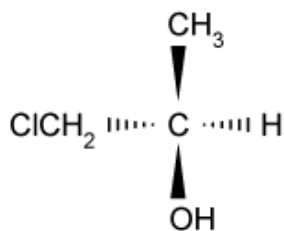
Compound 1

☐ A.



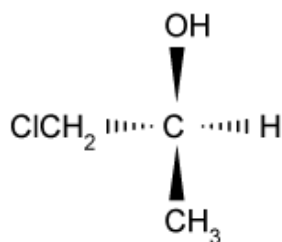
[10%]

✓ ☒ B.



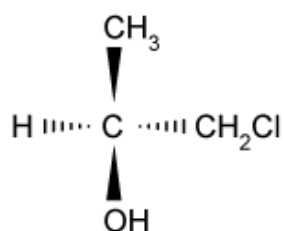
[78%]

☐ C.



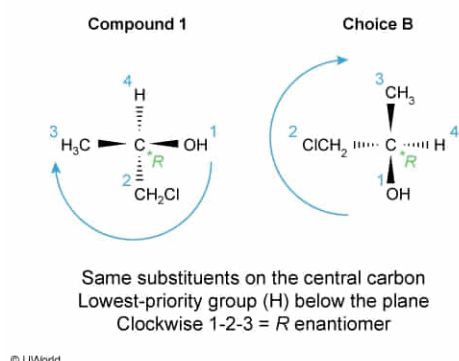
[5%]

☐ D.



Correct

78% answered correctly

Explanation:

To determine whether **chiral molecules** with the **same four substituents** on the central carbon are identical, the configuration of the **stereocenter** must be considered. First, the substituent **priority** is determined, and then *R* or *S* is assigned to each stereocenter. If the structure is shown with the lowest-priority group (usually hydrogen) pointing *away* from the viewer (dash) and priorities 1-2-3 follow a clockwise progression, the compound has an *R* configuration. (Priorities 1-2-3 following a counterclockwise progression indicate an *S* configuration.) Alternatively, if the structure is shown with the lowest-priority group pointing *toward* the viewer (wedge), the compound is being viewed from the opposite perspective. If priorities 1-2-3 follow a clockwise progression in this case, the compound has an *S* configuration. If two compounds with the **same four substituents** also have the **same stereocenter configuration**, they are the **same compound**.

The priority of the substituents on Compound 1 is (highest to lowest): OH, CH₂Cl, CH₃, and H. The lowest-priority group is away from the viewer, and priorities 1-2-3 follow a clockwise progression, indicating an *R* configuration. The correct answer also has an *R* configuration. It can also be **rotated 90°** to the left to put the substituents in the same position as Compound 1.

(Choices A, C, and D) These structures are **enantiomers of Compound 1**. They have the same four substituents but in an *S* configuration (opposite that of Compound 1). They are not identical to Compound 1, but are identical to one another.

Educational objective:

Chiral compounds with the same four substituents on the central carbon are identical if they have the same stereochemical configuration. When the lowest-priority substituent is pointing away from the viewer, an *R* configuration is present if the substituent priorities follow a clockwise progression, whereas a counterclockwise progression indicates an *S* configuration. If the lowest-priority substituent is pointing *toward* the viewer, an *R* configuration is indicated by substituent priorities following a *counterclockwise* progression.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401604

System:
Introduction to Organic Chemistry